

# LABORATORY RECORD

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Week No.:** 3

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## 1. TITLE

Write a Program Using JavaScript (Arrays, Functions)

## 2. AIM

To write JavaScript programs demonstrating array manipulation methods, custom functions, and ES6 class objects.

## 3. OBJECTIVE

The objectives of this experiment are:

- Practice JS array functions (filter, reduce).
- Implement ES6 class syntax.

## 4. THEORY

JavaScript provides array helper functions (filter, reduce, map) and ES6 class syntax for OOP.

## 5. ALGORITHM / PROCEDURE

1. Create function.js and object.js.
2. Implement score processing with filter and reduce.
3. Define Book class.
4. Run using Node.js.

## 6. PROGRAM

### function.js

```
// Student Score Processing in JavaScript
```

```
let scores = [78, 85, 62, 90, 94];
```

```
function printScores(arr) {  
    console.log("Scores:", arr);  
}
```

```
function calculateSum(arr) {  
    let sum = 0;  
    for (let i = 0; i < arr.length; i++) {  
        sum += arr[i];  
    }  
    return sum;  
}
```

```
function calculateAverage(arr) {
    return calculateSum(arr) / arr.length;
}

function addScore(arr, newScore) {
    arr.push(newScore);
}

printScores(scores);
console.log("Total Score:", calculateSum(scores));
console.log("Average Score:", calculateAverage(scores).toFixed(2));

console.log("\nAdding new score: 88");
addScore(scores, 88);
printScores(scores);
console.log("New Average Score:", calculateAverage(scores).toFixed(2));
```

### object.js

```
// Student Course System in JavaScript (ES6 Class)
```

```
console.log("==== Student Management System =====");
```

```
class CourseManager {
    constructor() {
        this.courses = [];
    }

    addCourse(title, credits) {
        this.courses.push({ name: title, points: credits });
        console.log(`Course "${title}" enrolled successfully.`);
    }

    listCourses() {
        console.log("\nEnrolled Courses:");
        for (let i = 0; i < this.courses.length; i++) {
            console.log(`${i + 1}. ${this.courses[i].name} (${this.courses[i].points} Credits)`);
        }
    }

    totalCredits() {
        let total = 0;
        for (let i = 0; i < this.courses.length; i++) {
            total += this.courses[i].points;
        }
        return total;
    }

    removeLastCourse() {
        let dropped = this.courses.pop();
        console.log(`\nDropped Course: ${dropped.name}`);
    }
}

const manager = new CourseManager();
manager.addCourse("Full Stack Web Dev", 4);
manager.addCourse("Machine Learning", 3);
manager.addCourse("Database Systems", 3);
```

```

manager.listCourses();
console.log("\nTotal Credits:", manager.totalCredits());

manager.removeLastCourse();
console.log("\nUpdated List:");
manager.listCourses();
console.log("Updated Total Credits:", manager.totalCredits());

```

## 7. CODE EXPLANATION

| Code / Syntax      | Explanation               |
|--------------------|---------------------------|
| scores.filter(...) | Filters array elements    |
| scores.reduce(...) | Aggregates array elements |
| class Book { ... } | ES6 OOP class             |

## 8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

| Test Case | Input / Action    | Expected Result                  | Actual Result                    | Status |
|-----------|-------------------|----------------------------------|----------------------------------|--------|
| TC-01     | Node.js Execution | Console prints processed outputs | Console prints processed outputs | Pass   |

## 9. OUTPUT

### Output 1: Execution Screenshot

```

Workspaces Applications Aug 30, 3:56 PM
~/H17week3
node week03.js
Courses: [ 80, 85, 90, 95, 98 ]
Total Score: 458
Average Score: 91.6
Adding new score: 88
Courses: [ 80, 85, 90, 92, 94, 98 ]
New Average Score: 92.83

```

## 10. RESULT

Thus, the experiment for Week 3 (Write a Program Using JavaScript (Arrays, Functions)) was successfully executed and verified.